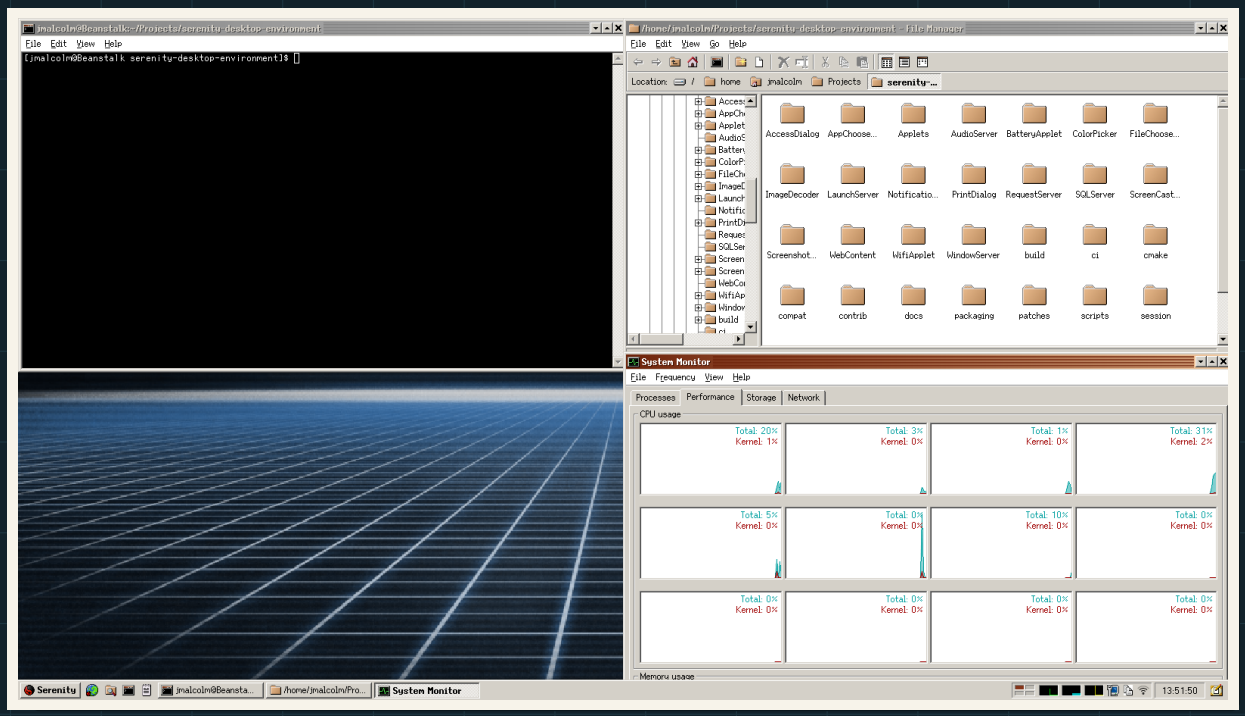
SDE · USER MANUAL

Serenity Desktop Environment

User Manual

The SerenityOS desktop experience on Linux

The SerenityOS Taskbar, applets, window chrome and applications, running as a normal Wayland session on Linux.



Taskbar and applets · Terminal · File Manager · System Monitor

October 2026 edition

for Linux · Wayland

About this manual

This manual is for people who use the Serenity Desktop Environment (SDE): a Linux desktop that recreates the SerenityOS experience on top of Wayland.

SDE is not a port of SerenityOS and it is not an emulator. It is a Linux desktop environment that runs the real SerenityOS shell and the real SerenityOS applications beside ordinary Linux applications. The kernel, the display server, the input stack and the services underneath are Linux's; the look, the widgets and the behaviour on top are SerenityOS's. This manual explains how to install SDE, how to use the desktop, and where the experience deliberately differs from SerenityOS itself.

Who this is for

You should be comfortable logging in to a Linux machine, opening a terminal, and editing a text file. No SerenityOS experience is assumed. Where a task needs the command line, the exact command is given.

How to read it

Chapters 1 to 3 get you installed and orient you on the desktop. Chapters 4 to 12 cover day-to-day use: windows, workspaces, the Taskbar and applets, applications, settings, Linux applications, screenshots, accessibility and input methods. Chapters 13 to 16 are reference material: the differences from SerenityOS, the session and recovery model, every configuration file and environment variable, and troubleshooting. The appendices are a shortcut card, a glossary and a list of further reading.

Conventions

Convention	Meaning
Super	The Super key: the Windows key on most keyboards, Command on Apple keyboards.
Ctrl+Alt+→	Press the keys together. A sequence such as **Super**, then **Tab** is written out in words.
`monospace`	A command, file name, path, configuration key or literal value.
Display Settings ▸ Themes	A menu path: open Display Settings, then the Themes tab.

> **SDE vs SerenityOS**
> A box like this marks something that behaves differently from SerenityOS, or something SDE adds that SerenityOS does not have. Chapter 13 collects all of the differences in one place.

> **Edition and status**
> This is the October 2026 edition. It describes the development state at milestone R6 (hardening and packaging). SDE is under active development: commands, file formats and behaviour can change between releases. The authoritative source for any one detail is the document set in the project's `docs/` directory, which this manual summarises for users.

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1 Welcome to SDE

SDE brings the SerenityOS desktop to Linux: the same Taskbar, the same window frames, the same applications and the same design language, running as a normal Wayland session on ordinary hardware.

1.1 What SDE is

SerenityOS is a from-scratch operating system with a distinctive late-1990s workstation aesthetic. Its desktop is not a set of configuration files on top of a toolkit: the widget toolkit (LibGUI), the graphics library (LibGfx), the applications, the shell and the window server all belong to the same system, and they were designed together. That coherence is the point.

SDE keeps that coherence and replaces the layer underneath it. The SerenityOS applications are used unmodified wherever possible. A compatibility library, LibWindowServer, speaks the applications' native WindowServer protocol on one side and Wayland on the other. A wlroots-based compositor, also called **WindowServer**, provides everything that used to be global inside SerenityOS: window management, the shell, the Taskbar, workspaces, display and input configuration, screen capture and the session lifecycle.

The result is a desktop that looks and behaves like SerenityOS but boots from a Linux login manager, drives DRM/KMS and libinput, uses PipeWire for audio, and can run GTK, Qt and X11 applications next to SerenityOS ones.

1.2 The design principles you will notice

Four ideas shape almost everything in the interface:

- **The shell is the real shell.** The Taskbar, the Desktop, the applets and the Settings applications are the actual SerenityOS programs. They are not a reimplementation with a similar look.
- **One design language.** Applications, dialogs and window frames share one theme, one font family and one set of widgets. A SerenityOS application and a GTK application side by side still look like they belong to the same desktop.
- **Classic window management.** Server-drawn title bars with Close, Maximize and Minimize buttons; a window menu; a taskbar; Alt+Tab; a grid of workspaces.
- **Power-user keyboard access.** Most window and workspace operations have a shortcut, and the whole set is configurable.

1.3 SDE and SerenityOS at a glance

If you have used SerenityOS, the desktop will feel immediately familiar. The differences are mostly underneath, plus a handful of deliberate additions. This table is the short version; Chapter 13 is the full account.

	SerenityOS	SDE
Kernel	SerenityOS kernel	Linux (x86-64)
Display	Built-in WindowServer over its own IPC	Wayland compositor (wlroots); apps reach it through LibWindowServer
Shell	Taskbar, Desktop, applets	The same programs, running as Wayland clients
Applications	SerenityOS userland	The same applications, built for Linux, with a few Linux-native exceptions

	SerenityOS	SDE
Services	SystemServer and Serenity services	A session of host services with the same protocols; no SystemServer
Login	LoginServer	Your display manager, or a TTY launcher
Other apps	None: the SerenityOS world only	Native Wayland and X11 applications through XWayland
Home directory	`/home/anon`	Your real Linux home; Serenity config is seeded into `~/.config`
Resources	`/res`, `/bin`	Redirected to the SDE installation; see §13.6

1.4 What you need

Requirement	Details
Computer	x86-64, with a GPU supported by Linux DRM/KMS and Mesa. A software renderer (pixman) is available as a fallback.
Linux	A modern Linux with Wayland support. The packaged path targets Arch Linux and EndeavourOS.
Graphics session	A DRM device (`/dev/dri/card*`), or a nested session inside another Wayland compositor for development.
Seat	logind (systemd) or seatd, so the compositor can take DRM master and switch virtual terminals. A *builtin* seat backend exists for simple setups.
Audio	PipeWire, for the audio service and the Audio applet.
Optional	NetworkManager for the Wi-Fi applet; `swaylock` or another `ext-session-lock-v1` locker; `xdg-desktop-portal` for portal-aware applications; `at-spi2-core` for screen readers.

> **This is young software**
> SDE is an ambitious compatibility layer, not a finished consumer product. Some corners are still being built, and the packaging has not yet been validated end-to-end on a clean machine. Keep a second, known-good session installed and a spare virtual terminal available. Chapter 16 explains the escape hatches.

1.5 How this manual is organised

Chapters 1–3	Installing SDE, starting a session, and a guided tour of the desktop.
Chapters 4–8	Everyday use: windows, workspaces, shortcuts, the Taskbar and applets, and applications.
Chapters 9–12	Settings and appearance, Linux applications and interoperability, screenshots and screen sharing, accessibility and input methods.
Chapters 13–16	Reference: differences from SerenityOS, the session and recovery model, configuration, troubleshooting.
Appendices	Shortcut card, glossary and further reading.

2 Installing and starting

SDE installs as a Wayland session that your login manager offers alongside your other desktops. This chapter covers both the packaged install and the developer install from a checkout, then explains what starts when you log in.

2.1 Before you install

You need a working graphical Linux system already, because you install SDE from it and can fall back to it. Check three things first:

1. **Your GPU works.** `/dev/dri/card0` should exist and your normal session should be able to use it. SDE can fall back to software rendering, but the GPU path is the intended one.
2. **You have a seat provider.** On systemd systems, logind is already there. Without systemd, install `seatd` and make sure your user is in the `video` and `input` groups if you use the builtin backend.
3. **You have a lock client.** `Super+L` runs `swaylock` by default. Without a locker, locking does nothing. Install `swaylock` or set `lock_command` in `session.conf` (Chapter 15).

> **Optional components**
> The Wi-Fi applet needs NetworkManager running on the system bus. The battery applet reads `/sys/class/power_supply` directly and needs nothing extra. Screen readers need `at-spi2-core`. Portal-aware applications need `xdg-desktop-portal` plus the SDE backend (Chapter 10).

2.2 Installing the packages

SDE is split into three Arch packages under `packaging/aur/`:

Package	What it provides
`serenity-lagom-git`	The LibGUI/LibGfx framework, LibWindowServer, the SerenityOS applications, resources, fonts and the skeleton user configuration. Installed under `/usr/lib/sde`.
`serenity-desktop-environment-git`	The compositor, session services, applets, portal dialogs, tools and the session entry. Installed under `/usr/lib/sde` and `/usr/share/wayland-sessions`.
`wlroots0.20`	libwlroots 0.20.2 and its development files, installed under `/usr`. The SDE package also bundles the wlroots libraries it was built against.

> **The AUR packages are not published yet**
> The three PKGBUILDs are prepared but have not been uploaded, and a clean package install followed by a login has not yet been confirmed. Build them locally from the repository:

```
git clone https://github.com/jmalcolm137/serenity-desktop-environment
cd serenity-desktop-environment/packaging/aur
(cd wlroots0.20 && makepkg -si)
(cd serenity-lagom-git && makepkg -si)
(cd serenity-desktop-environment-git && makepkg -si)
```

The build compiles the SerenityOS Lagom framework and the application suite, which is large: expect tens of gigabytes of build tree and a long compile. After installation, log out and choose **Serenity Desktop Environment** at the login screen.

Developer install from a checkout

If you are working on SDE itself, build the tree and install only the session entry. The launcher runs the checkout directly, so a rebuild is all that is needed to pick up changes:

```
scripts/build-full-sde.sh --with-apps
sudo scripts/install-session.sh          # /usr/share/wayland-sessions/sde.desktop
sudo scripts/install-session.sh --uninstall
```

This writes `/usr/local/bin/sde-session`, which execs `scripts/start-sde-session.sh` in this checkout. Rebuild the compositor before logging in (`scripts/build-window-server.sh`).

2.3 Starting a session

How	Command or action	When to use it
Display manager	Choose **Serenity Desktop Environment** at the login screen.	Normal use. The session is registered with logind, so DRM master and VT switching are handled for you.
TTY	`scripts/start-sde-session.sh`	Development, or when no display manager is running. `XDG_RUNTIME_DIR` must be set.
Nested	`scripts/start-sde-nested.sh`	Development inside another Wayland session. The SDE session appears in a window on the host desktop.

The launcher refuses to start if it finds no DRM device, because that almost always means a seat or permissions problem rather than a missing GPU. The message suggests the fix. For a deliberately headless run (CI, testing), set `SDE_BACKENDS=headless`.

2.4 What starts with the session

There is no SystemServer in SDE. The session script starts the supervised services, the compositor and the shell, then your autostart entries and any restored applications.

Started	What it is
WindowServer	The compositor. It creates the Wayland socket `sde` (or `sde-<seat>`) and the window-manager socket `/tmp/portal/wm`.
Taskbar	The SerenityOS Taskbar, as a layer-shell panel. It hosts the window list, applets and clock.
Desktop	`FileManager -d`: the SerenityOS Desktop, showing the contents of your desktop directory as icons over the wallpaper.
Applets	Workspace picker, CPU/memory/network graphs, network, clipboard history, audio, keymap, plus SDE's Wi-Fi and battery applets.
LaunchServer, ImageDecoder	Supervised session services. They serve the SerenityOS protocols for application launching and image decoding.
Portal backend	`xdg-desktop-portal-serenity`, unless `SDE_SESSION_PORTAL=0`.
Idle and sleep helpers	`sde-idle` (blank, then lock) and `sde-sleep.sh` (lock on suspend, unblank on resume).
Autostart entries	Everything in `$XDG_CONFIG_HOME/autostart/*.desktop` that is not hidden.
Restored apps	The applications recorded in `~/.config/sde/session-apps` from your last session.

2.5 Logging out, restarting and shutting down

Open the system menu (**Super** tapped on its own, or the start button) and choose **Exit…**. The dialog offers three actions:

- **Power off computer** — shuts the machine down through logind.
- **Reboot** — restarts the machine.
- **Log out** — ends the SDE session and returns to the login screen.

All three go through the session's power helper, so they also work when systemd is not present, provided a suitable fallback command exists. From a terminal, the same operations are available as `sde-power.sh logout|reboot|shutdown|suspend`. A display manager's logout signal is forwarded to the whole session, so the compositor and its services stop cleanly.

2.6 Your first five minutes

1. Tap **Super** to open the system menu. Everything installed is in the category submenus; **Themes** changes the look of every window immediately.
2. Choose **Utilities ▸ Terminal**. Run `uname -a` and `fastfetch` if it is installed. This is a normal Linux shell: you can run `htop`, `vim` or a Linux GUI program from it.
3. Choose **Utilities ▸ File Manager**, or double-click a folder or file on the Desktop. Your real home directory is there.
4. Open **Settings ▸ Display Settings** and look at the six tabs. **Monitor** changes resolution and scale live; **Themes** changes the whole desktop; **Workspaces** sets the size of the workspace grid.
5. Hold **Alt** and press **Tab** to switch windows. Release **Alt** to activate the selection. Use **Ctrl+Alt** with the arrow keys to move around the workspace grid.
6. Press **Super+L** to lock the screen. If nothing happens, install a lock client (see §2.1).

3 A tour of the desktop

The SDE desktop is made of four things: the wallpaper and desktop icons, the Taskbar along the bottom, the windows themselves, and the system menu behind the start button.

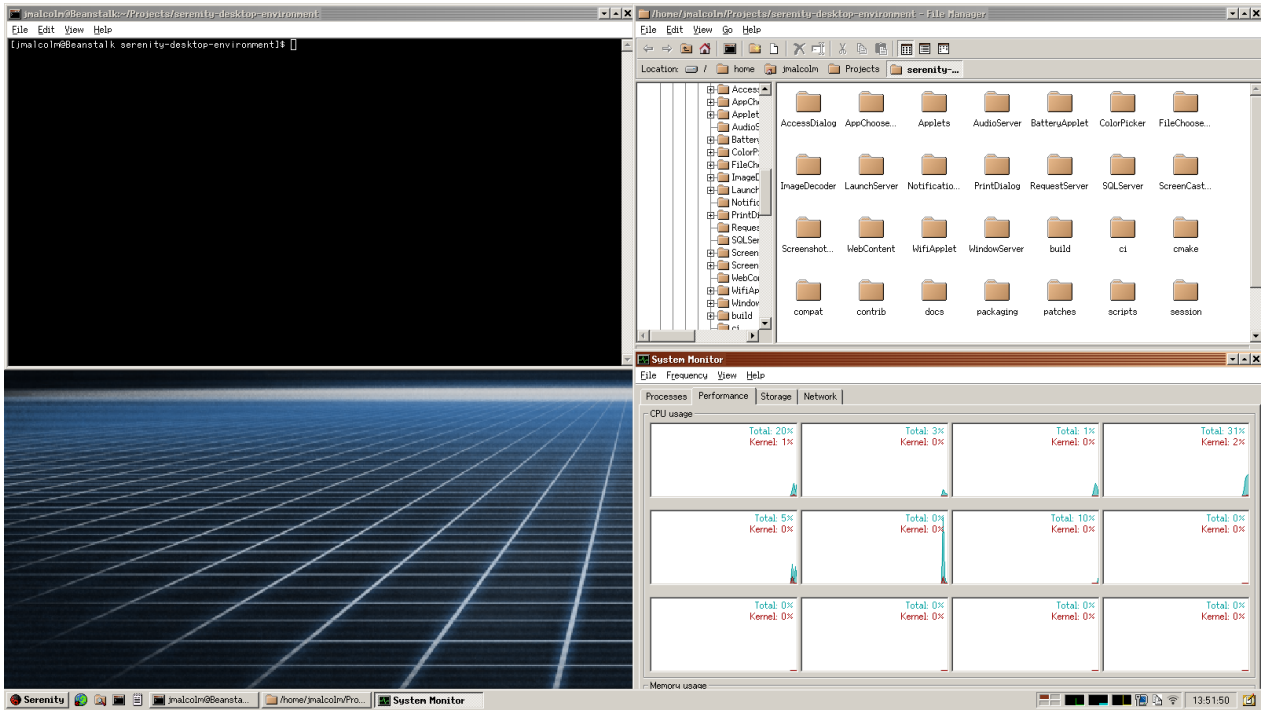


Figure 1. *An SDE session: the SerenityOS Taskbar and its applets along the bottom, a Terminal and the File Manager in SerenityOS frames, and System Monitor. The Desktop itself is the File Manager in desktop mode, which shows your desktop directory as icons over the wallpaper (not shown here; see §3.1).*

3.1 The desktop

The Desktop is the SerenityOS File Manager in desktop mode. It shows the contents of your desktop directory (normally `~/Desktop`) as icons over the wallpaper. Double-click an icon to open it, drag icons to arrange them, and right-click the background for a menu with **New Directory**, **New File**, **Paste**, **Open in Terminal** and **Display Settings**.

The wallpaper is painted by the compositor, so it covers the whole output before any window appears, including across multiple monitors. Set it in **Display Settings ▸ Background**, or right-click an image in File Manager and choose **Set as Wallpaper** (the upstream SerenityOS behaviour).

3.2 The Taskbar

The Taskbar is the SerenityOS Taskbar, with two small SDE patches (a Launch item in the quick-launch menu and optional window grouping). It is a layer-shell panel, so it reserves space at the bottom of the screen: maximised windows stop above it rather than covering it. From left to right:

Part	What it does
Start button	The SerenityOS ladyball button. Left-click opens the system menu; a **Super** tap does the same.
Quick launch	A row of pinned applications. Right-click an icon for the application menu; in SDE it gains a **Launch** item above **Remove**.
Window list	One button per open window, in stacking order. Click to focus or minimise; right-click for the window menu. Buttons show the window icon and title.

Part	What it does
Applet area	Workspace picker, resource graphs, network, clipboard history, audio, keymap, Wi-Fi and battery. See §3.5.
Clock	The time. Click it to open a calendar popup; the popup has a button to launch the Calendar application.
Show desktop	Minimises every window, or restores them if the desktop is already showing. Also available as **Super+D**.



*Figure 2. The Taskbar in two halves, with the parts numbered: **1** start button, **2** quick launch, **3** window buttons, **4** applet area, **5** clock. The show-desktop button is at the far right, after the clock. The exact applet set and order are configurable (§7.13).*

3.3 The system menu

The system menu is the SerenityOS application menu. It is generated at startup from the installed `.af` application files, so it always reflects what is actually installed.

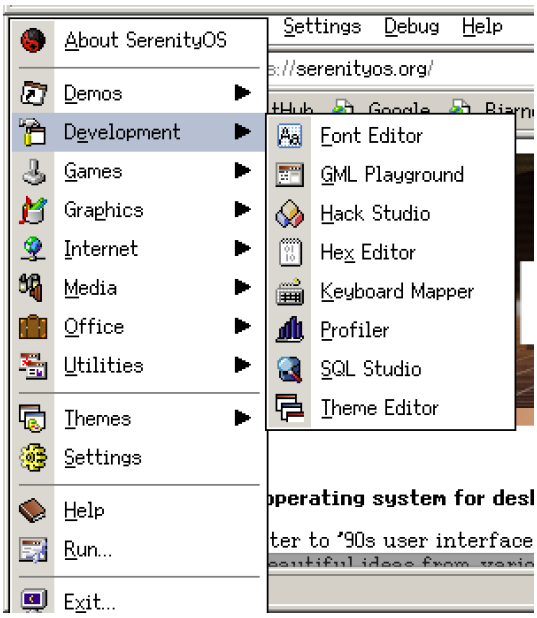


*Figure 3. The system menu with the **Development** submenu open. The categories, the Themes submenu and the Settings, Help, Run and Exit items are all part of the SerenityOS Taskbar.*

Item	What it does
About SerenityOS	Opens the About application, with the SerenityOS userland version and credits.
Demos, Development, Games, Graphics, Internet, Media, Office, Utilities	Application categories. Each submenu lists the installed programs in that category.
Themes	A checkable list of every installed theme. Choosing one applies it to the whole desktop and every application.
Settings	Opens the Settings application: a launcher for the individual settings programs.
Help	Opens Help, which browses the installed manual pages.

Item	What it does
Run…	A small dialog for launching a program by path or name, in your home directory.
Exit…	Power off, reboot or log out (see §2.5).

The menu is a normal popup menu: hover a submenu to open it, click outside to dismiss it, and use the arrow keys and **Enter** if you prefer the keyboard. The system menu can be opened with a **Super** tap only when you have not used another key or mouse button while Super was held.

3.4 The window list

Every top-level window gets a Taskbar button. The button is raised for the active window; minimised windows show a pressed-in button. Hovering a button shows a tooltip with the window title. Clicking the active window's button minimises it; clicking a minimised window's button restores it.

By default SDE keeps SerenityOS's one-button-per-window model. If you prefer the grouped model used by many Linux desktops, set `[Taskbar] GroupWindows=true` in `WindowServer.ini`. Windows of the same application then share one button, and clicking an active group cycles through its windows.

> **Grouping is an SDE option**
> SerenityOS always shows one Taskbar button per window. Grouping by application is an opt-in SDE addition, off by default.

3.5 The applet area

Applets are small status and control widgets that live at the right end of the Taskbar. SDE starts the SerenityOS applets and adds two of its own (Wi-Fi and battery). The order is configurable with `[Applet] Order`; applets are laid out right-to-left, so the last entry is the rightmost.

Applet	What it shows	What it does
Workspace picker	The workspace grid (2×2 by default).	Click a cell to switch. Wheel changes the row, Shift+wheel the column.
Resource graphs	Three small live graphs: CPU, memory and network.	Click opens System Monitor on the matching page.
Network	Link state; a tooltip lists interfaces with addresses and speeds.	Click opens System Monitor's Network page.
Clipboard history	A clipboard icon.	Click for the history window; right-click an entry to delete, dump or clear.
Audio	Speaker icon, percentage or mute.	Click for the volume slider, right-click to mute, wheel to change volume.
Keymap	The active keyboard layout, e.g. `us`.	Shows the layout; use **Alt+Shift** to switch (see §12.6).
Wi-Fi	Signal strength, disconnected or off.	Click to scan, connect and disconnect networks. SDE addition; needs NetworkManager.
Battery	Charge level with a charging bolt, or a slash if no battery.	Click for per-battery details and time remaining; right-click to toggle the percentage.

Most applets have a tooltip with more detail, and most open a popup on click. Popups are dismissed by clicking the applet again; the battery and Wi-Fi popups do not yet close on an outside click (§16.2).

3.6 The clock and the calendar

The clock shows the time in the system time zone. Click it for a month calendar with previous/next month buttons and a jump-to-today button; the calendar button opens the Calendar application. The clock's tooltip shows today's date. The time format (12/24-hour) and the time zone are set in **Clock Settings**.

3.7 Show desktop

The button at the far right minimises every window and shows the wallpaper and desktop icons. Press it again (or **Super+D**) to restore the windows. The Taskbar's own right-click menu offers the same action, with the label changing to **Show open Windows** when the desktop is already visible.

3.8 Notifications

Applications can post notifications through the SerenityOS notification service. They appear as borderless banners in the top-right corner, above all windows, and are not part of the window list or Alt+Tab. Hovering expands a notification; clicking it opens the action's target.

> **Notifications do not time out**
> In this release a notification stays until you click it or the application withdraws it. SerenityOS's server in this snapshot has no auto-dismiss timer, and SDE has not added one.

3.9 The Taskbar's own menu

Right-clicking an empty part of the Taskbar opens a short menu with **Open System Monitor**, **Display Settings** and **Show Desktop**. This is a quick way to reach the two most-used settings without the system menu.

4 Windows

Windows in SDE are framed by the compositor, in the SerenityOS style: a textured title bar, an icon at the left and Close, Maximize and Minimize at the right. The behaviour follows SerenityOS, with a few SDE additions.



1 Window icon — click for the window menu 2 Title — drag to move; double-click to maximize
3 Minimize 4 Maximize / Restore 5 Close

Figure 4. *The parts of a SerenityOS window frame. The frame is drawn by the compositor, not by the application, so every window gets the same chrome.*

4.1 Title-bar buttons

Button	Action
Close (right)	Closes the window. Applications are asked to close and may prompt to save.
Maximize / Restore	Toggles between maximized and its previous size. The icon shows what the next click will do.
Minimize	Hides the window; its Taskbar button stays, pressed in. Click the button to restore it.
Window icon (left)	Left-click opens the window menu. Right-clicking the title bar does the same.

A button acts on release, and only if the pointer is still over it, so you can slide off a button to cancel. The title bar itself is a drag handle: hold the left button and move to reposition the window. Double-clicking the title bar maximizes the window, or restores it if it is already maximized; the two clicks must land on the same window within the double-click time (250 ms by default) and within a few pixels of each other.

Dragging a maximized window restores its previous size and keeps the pointer at the same relative position in the window, so the title bar does not jump away from the cursor.

4.2 Moving and resizing

Action	How
Move	Drag the title bar. Or hold **Super** and drag anywhere in the window.
Resize	Drag a window border or corner. The border is 4 px wide and shows a resize cursor. Or hold **Super** and drag with the right mouse button: the window is divided into a 3×3 grid and the cell you start in chooses the edge or corner.
Size readout	While moving or resizing, a small overlay shows the window's size and position. Turn it off with `[Effects] WindowGeometry=false`.
Snap to an edge	While moving a window, bring the pointer within 16 px of a screen edge. A translucent preview appears; release to snap. Eight targets are available: four halves and four quarters.

Drag-to-edge snapping is on by default. The top edge gives the top half of the screen, not maximize. To tune it, set `[Tiling] EdgeThreshold` (in pixels) or disable it with `[Tiling] DragToEdge=false`.

> **Snap by dragging is an SDE addition**
> SerenityOS tiles only through keyboard shortcuts. Dragging a window to a screen edge to snap it is an SDE extension.

4.3 The window menu

Open the window menu by clicking the window icon at the left of the title bar, by right-clicking the title bar, or by right-clicking the window's Taskbar button. It can also be navigated entirely from the keyboard: **Esc** closes it, the up and down arrows move, and **Enter** activates the highlighted item.

Item	What it does
Minimize / Unminimize	Toggles the minimized state.
Maximize / Restore	Toggles the maximized state.
Move	Starts an interactive move. Disabled while minimized or maximized.
Roll Up	Hides the window contents and leaves only the title bar. The window can still be moved. Maximizing a rolled-up window unrolls it first.
Menu Bar	Shows or hides the application's menu bar, where the application supports it.
Always on Top	Keeps the window above normal windows, including when focus changes.
Close	Closes the window. This is the default item, shown in bold.

The checkbox items (Roll Up, Menu Bar, Always on Top) show a tick when active. The menu closes when you choose an item, press **Esc**, click outside it, or click with a button other than the left one.

4.4 Maximize, minimize and fullscreen

Maximizing fills the usable area: the screen minus the Taskbar. The compositor remembers the previous geometry and restores it on the next maximize toggle. Minimized windows animate towards their Taskbar button; the animation can be disabled with `[Effects] AnimateWindows=false`. Fullscreen is different: a window goes fullscreen only when the application asks for it, and it then has no frame at all. Applications that play video or games normally offer this through their own menus or F11.

4.5 Focus

By default, focus follows clicks: clicking anywhere in a window raises it and gives it the keyboard. If you prefer the focus to follow the pointer, set `[Focus] FollowsMouse=true`. The change is applied live. New windows open on the current workspace, cascade from the previous window, and dialogs open centred.

SDE supports explicit placement rules. Add a `[Placement] Windows` entry to `WindowServer.ini` and the first rule whose title is a substring of a new window's title wins:

```
[Placement]
Windows=Terminal:0,0,640,400; File Manager:640,0,640,400
```

This is mainly useful for reproducible layouts and screenshots; most people will never need it.

4.6 Dialogs and modals

Application dialogs are ordinary windows with a SerenityOS frame. A modal dialog blocks its parent application until it is answered, and transient dialogs stay above the window that opened them. Linux applications that use client-side dialogs behave the same way within their own application.

One limitation remains: a modal dialog belonging to an X11 application does not block the other X11 windows at the window-manager level, and the same is true for cross-kind (X11 to Wayland) modal chains. SerenityOS's explicit “lower the blockers” pass has not been reproduced yet. In practice this is only noticeable in applications with several top-level windows.

4.7 Applications that draw their own title bar

Modern GTK and Qt applications may ask to draw their own title bar and buttons (client-side decorations). SDE honours the request: such a window gets no SerenityOS frame and uses its own buttons and drag behaviour. The compositor still handles their maximize, minimize and fullscreen requests. Native Wayland windows without client-side decorations get a SerenityOS frame.

4.8 X11 windows

X11 applications run through XWayland and get SerenityOS frames too. Most behaviour is the same: drag the title bar, resize from the border, use the three title-bar buttons, and use **Super**+drag to move or resize. Two differences are worth knowing:

- There is no window menu and no double-click-to-maximize on X11 title bars yet.
- The keyboard tiling and maximize actions (**Super**+arrows) act on native Wayland windows only. For an X11 window, use its title-bar buttons, **Super**+drag, or the application's own shortcuts.

X11 windows also take part in the Taskbar, Alt+Tab, workspaces, clipboard and drag selection. Chapter 10 covers the rest of the interoperability story.

5 Workspaces and switching

SDE arranges workspaces in a grid, exactly as SerenityOS does. By default the grid is 2×2; it can be anything from 1×1 up to 16×16.

5.1 The workspace grid

Each cell of the grid is an independent workspace with its own set of windows. The Taskbar, the applets and the wallpaper are shared: they are part of the shell, not of a workspace. New windows open on the workspace you are currently looking at, whether they are Wayland or X11 windows.

Switching workspaces is animated: the grid slides sideways, and a small overlay shows the full grid with the current cell highlighted for half a second. At the edge of the grid the overlay still appears, but the workspace does not change: the grid does not wrap around.

5.2 Moving between workspaces

Action	Shortcut
Switch to the workspace left, right, up or down	**Ctrl+Alt + arrow key**
Carry the focused window with you	**Ctrl+Shift+Alt** + the same arrow
Jump to a workspace	Click its cell in the Workspace picker applet

The Workspace picker applet shows the grid at all times. Clicking a cell switches to it; the mouse wheel changes the selected row and Shift+wheel changes the column (useful when the grid is larger than the applet). Right-clicking the applet offers **Workspace Settings**, which opens Display Settings at its Workspaces tab.

5.3 The window switcher

There are two switchers, and the difference matters:

Keys	Scope	Behaviour
Alt+Tab	Windows on the current workspace	Shows a strip of live thumbnails. Each **Tab** moves forward, **Shift+Tab** back. Release **Alt** to activate the selection.
Super+Tab	Windows on every workspace	The same strip, but over all workspaces. The visible workspace follows the selection as you cycle, so you see the window you are about to choose.

Pressing any other key while the switcher is open cancels it; so does **Esc**. The switcher is drawn by the compositor and is not part of the window list. It shows live thumbnails, not icons, so it can be used to identify a document by its contents.

5.4 Configuring the grid

Set the grid in **Display Settings ▸ Workspaces**, or in `WindowServer.ini`:

```
[Workspaces]
Rows=2
Columns=2
```

The values are validated (1 to 16) and applied live: the Workspace picker and the switching overlay re-grid without a restart.

6 Keyboard shortcuts

Most window and workspace operations have a keyboard shortcut. The tiling, workspace and lock bindings are configurable; the switcher and the Taskbar-related keys are fixed.

6.1 Configurable bindings

These live under `[Keybindings]` in `WindowServer.ini` and take effect immediately when the file changes.

Action	Default	What it does
`SwitchWorkspaceLeft/Right/Up/Down`	**Ctrl+Alt+←/→/↑/↓**	Move to the neighbouring workspace.
`CarryWorkspaceLeft/Right/Up/Down`	**Ctrl+Shift+Alt+←/→/↑/↓**	Move to the neighbouring workspace and take the focused window.
`TileLeft` / `TileRight`	**Super+←** / **Super+→**	Tile the window to the left or right half. Press again to restore.
`Maximize` / `Minimize`	**Super+↑** / **Super+↓**	Maximize/restore or minimize the focused window.
`MaximizeHorizontal`	**Super+Alt+←** or **Super+Alt+→**	Horizontal maximize tile; toggles.
`MaximizeVertical`	**Super+Alt+↑** or **Super+Alt+↓**	Vertical maximize tile; toggles.
`Lock`	**Super+L**	Starts the screen locker (`swaylock` by default).

The tile, maximize and carry actions operate on native Wayland windows. For an X11 window, use **Super**+drag or the title-bar buttons. A combination must include at least one modifier; an invalid binding is rejected, the previous value is kept, and the reason is logged.

6.2 Fixed shortcuts

Keys	Action
Alt+Tab / **Shift+Alt+Tab**	Window switcher for the current workspace; forward or backward.
Super+Tab	Window switcher across all workspaces.
Super (tap and release)	Opens the system menu, if no other key or mouse button was used in between.
Super+Space	Launches Assistant, the application and command launcher.
Super+D	Shows the desktop, or restores the windows.
Super+1 … **Super+9**	Activates the corresponding Taskbar window button, counting from the left.
Super+H	Toggles the cursor highlight: a coloured circle around the pointer. Useful in presentations and for low vision.
Ctrl+Super+Shift+I	Reloads icon and chrome bitmaps after a display-scale change.
Alt+Shift	Switches to the next keyboard layout, when more than one is configured.
Alt+Escape	Quits the compositor and ends the session immediately. See the warning below.

> **Alt+Escape ends the session without asking**
> This is a development convenience inherited from the compositor and it is not configurable. Any combination containing Alt+Escape (including Ctrl+Alt+Escape) quits the compositor at once. Running applications are terminated; if session restore is enabled, they are relaunched on the next login.

The **Super+Space**, **Super+D** and **Super+digit** shortcuts are forwarded to the Taskbar. They only fire if **Super** was freshly pressed; if you hold Super and press several digits, only the first is forwarded.

6.3 Keys inside the window menu

When a window menu is open it takes the whole keyboard: **Esc** closes it, the up and down arrows move, and **Enter** activates the highlighted item. All other keys are ignored, including the underlined accelerator letters that are drawn in the menu.

6.4 Key repeat and shortcut inhibition

Key repeat is implemented by the compositor, with a 600 ms delay and 25 repeats per second by default. Both values are configurable in `input.ini`. Keys used as shortcuts do not repeat. An application that grabs the keyboard with the `keyboard-shortcuts-inhibit` protocol, such as a virtual machine viewer or a game, suspends all compositor shortcuts while it has focus, including **Alt+Tab** and **Super**.

6.5 Changing a binding

Add or edit an action under `[Keybindings]`. The value is one or more comma-separated combinations, each made of modifiers and an XKB keysym name. Modifiers are **Ctrl**, **Alt**, **Shift** and **Super** (aliases **Logo** and **Meta**).

```
[Keybindings]
Lock=Super+Escape
TileLeft=Super+Left, Ctrl+Alt+Left
SwitchWorkspaceLeft=Ctrl+Alt+Page_Up
```

Changes are validated and applied live. You can check the effective result from a checkout with `sde-config validate` and `sde-config list`, or ask the compositor directly with `WindowServer --print-config`. In a packaged install, edit the file and watch the session log if a value is rejected.

> **Terminal keys**
> Inside the Terminal, copy and paste are **Ctrl+Shift+C** and **Ctrl+Shift+V**, so that **Ctrl+C** continues to mean interrupt. **Ctrl+letter** sends the corresponding control character, **Alt+key** sends Escape followed by the key, and the Terminal passes the host environment through, so you can start Linux GUI programs from it.

7 The Taskbar and its applets

The Taskbar is the centre of the session: it launches applications, lists windows, hosts the applets and shows the clock. Almost everything here is the SerenityOS Taskbar, plus two SDE applets for Wi-Fi and battery.

7.1 Starting applications

There are four common ways to start a program:

- **The system menu** (start button or a **Super** tap): browse by category, or use the Themes, Settings, Help and Run items.
- **Quick launch**: click a pinned icon. Right-click the icon to open the application's own menu; in SDE the quick-launch context menu also offers **Launch** above **Remove**.
- **Run…** in the system menu, or **Super+Space** for Assistant: type a program name or a path.
- **A Terminal**: type the command. GUI programs started here appear as normal SDE windows.

7.2 The window list

Each open window has a button showing its icon and title. The active window's button is raised; a minimised window's button is pressed in. Buttons are ordered by stacking, so the most recently focused windows are easiest to find. Right-click a button for the window menu, or left-click the active button to minimise.

Grouping. With `[Taskbar] GroupWindows=true`, buttons are grouped per application (per Wayland `app_id` or X11 `WM_CLASS`). Clicking the active group cycles through its windows; clicking a group with one window still toggles minimise.

7.3 Workspace picker

The leftmost applet is the workspace picker. It draws the grid and highlights the current cell. Click a cell to jump; the mouse wheel moves the selection through rows, and Shift+wheel through columns. Right-click and choose **Workspace Settings** to open the Workspaces tab of Display Settings. Chapter 5 covers the shortcuts.

7.4 Resource graphs

Three small graphs show CPU, memory and network activity. On SerenityOS they read the kernel's own statistics; in SDE they read `/proc/stat`, `/proc/meminfo` and `/sys/class/net/*/statistics`. Click a graph to open System Monitor on the matching page. Their tooltips give the current value.

7.5 Network applet

The network applet shows whether an interface is connected. Its tooltip lists interfaces with their IPv4 address and link speed, or says that an interface is down. Clicking it opens System Monitor's Network page. It reports IPv4 only, and the optional connect/disconnect notifications are not enabled by the session.

7.6 Clipboard history

The clipboard applet keeps a history of copied text. Click it to open the history window; click an entry to copy it again. Right-click an entry to **Delete** it, **Dump to Debug Console**, or **Clear History**. The history is shared with the clipboard used by native Wayland and X11 applications, so copying in a GTK application and pasting into a SerenityOS one works.

7.7 Audio

The speaker applet shows the current volume or the word **mute**. Click it for a slider; right-click toggles mute; the mouse wheel changes the volume while unmuted. The volume it controls is the main output of SDE's audio service, which is backed by PipeWire. The Piano and SoundPlayer applications use the same service. Applications that use their own PulseAudio/PipeWire output are independent of it.

7.8 Keymap

The keymap applet displays the active keyboard layout (for example `us`). Its menu can open Keyboard Settings, but choosing a layout from the applet does not switch it in this release: the compositor's keymap request is still a stub. Use **Alt+Shift** to cycle the layouts configured in `input.ini`, or Keyboard Settings to change them.

> **Keymap switching**
> SerenityOS switches layout from the applet. In SDE, use Alt+Shift or the Keyboard Settings application; the applet itself only reports the layout.

7.9 Wi-Fi

The Wi-Fi applet is an SDE addition: a LibGUI front end for NetworkManager, modelled on KDE Plasma's Networks applet. It uses the system D-Bus, so NetworkManager must be running; without it the applet reports that networking is unavailable.

Interaction	What happens
Click the applet	Opens the popup: a Wi-Fi on/off switch, a search field, **Hidden…**, **Configure** and the list of visible networks.
Click a network	Connects to it. A secured network asks for the passphrase inline; **Enter** connects, **Esc** cancels.
Click the connected network	Disconnects.
Right-click the applet	**Enable Wi-Fi**, **Connect to Hidden Network…**, **Configure Network Connections…**.

The icon shows signal strength, a red cross for disconnected, a red slash for Wi-Fi off and a yellow spinner while connecting. Saved networks connect automatically; the applet also answers NetworkManager's passphrase requests for saved profiles. **Configure** runs `nm-connection-editor` if it is installed.

Limitations to be aware of: there is no VPN or mobile-broadband support; all security types are shown with the same lock icon; a captive-portal state is displayed but not resolved; and WPA-Enterprise (802.1X) is not supported from this applet. Use `nmcli` or `nm-connection-editor` for those.

7.10 Battery

The battery applet reads `/sys/class/power_supply` directly; it does not use UPower or D-Bus. The icon is green normally, orange at 20% or below and red at 5% or below, with a bolt while charging and a slash when the machine has no battery. Click it for one row per battery (name, model, bar, percentage and time remaining or until full); right-click toggles the percentage display.

It shows the system batteries only: there is no Bluetooth-device battery support and no brightness slider. The popup does not close when you click outside it; click the applet again.

7.11 Clock

The clock shows the time and, on hover, today's date. Click it for a calendar popup with previous/next month, a jump-to-today button and a button to open the Calendar application. Time format and time zone are set in Clock Settings.

7.12 Show desktop

The far-right button minimises all windows; when the desktop is showing, it restores them. The same action is available as **Super+D** and from the Taskbar's right-click menu.

7.13 Choosing and ordering applets

Set `[Applet] Order` in `WindowServer.ini` to a comma-separated list. Applets are placed right-to-left, so the *last* entry is the rightmost. SDE appends **WiFi** and **Battery** to the default list, which puts Wi-Fi just left of the battery at the far right.

```
[Applet]
Order=WorkspacePicker,CPUGraph,MemoryGraph,NetworkGraph,Network,ClipboardHistory,Audio,Keymap,WiFi,Battery
```

Removing a name hides that applet; it does not uninstall it. Changes are applied live. The clock and the show-desktop button are part of the Taskbar itself and are always present.

8 Applications

SDE ships the SerenityOS application suite: largely the same programs, menus and dialogs as upstream, built for Linux (a few are Linux-native adaptations). More than seventy applications are built, covering everyday tools, settings, games, demos and developer tools.

8.1 What is included

Category	Applications
Everyday	Terminal, File Manager, Text Editor, Calculator, Image Viewer, PDF Viewer, 3D File Viewer, Sound Player, Video Player, Calendar, Clock, Analog Clock, Character Map, Help, Assistant, Space Analyzer, Spreadsheet, Piano, Weather, Maps, Mail, Presenter, Hex Editor, Font Editor
Internet	Browser, Mail, Maps, Weather, Assistant, Help
Settings	Settings, Display Settings, Mouse Settings, Keyboard Settings, Theme Editor, Terminal Settings, Network Settings, Calendar Settings, Clock Settings, Mail Settings, Browser Settings, Maps Settings, Games Settings, Certificate Settings, Users Settings
Games	2048, BrickGame, Chess, ColorLines, FlappyBug, Flood, Game of Life, Hearts, MasterWord, Minesweeper, Snake, Solitaire, Spider, Sudoku
Development	Hack Studio, Profiler, Debugger, Crash Reporter, SQL Studio, GML Playground, Font Editor, Keyboard Mapper, Hex Editor, System Monitor
Demos and utilities	CatDog, Eyes, Gradient, LibGfx demos, Mandelbrot, Model Gallery, Screensaver, Starfield, Widget Gallery, Tubes, Magnifier, Screenshot

The system menu is generated from the installed application files, so this list and the menu stay in step. A few programs that exist in the SerenityOS tree are not built by default: **PixelPaint**, **PartitionEditor** and **Welcome**.

8.2 Terminal

Terminal is a full VT with scrollback, colour, mouse reporting and the usual escape sequences. It runs your login shell in your real home directory. The shell environment is passed through from the session, including `PATH`, `HOME`, `LANG`, `WAYLAND_DISPLAY` and `DISPLAY`, so Linux command-line and GUI programs behave normally when started from it.

Keys	Action
Ctrl+Shift+C / **Ctrl+Shift+V**	Copy selected text / paste.
Ctrl+C	Send SIGINT (interrupt the foreground program).
Ctrl+letter	Send the corresponding control character.
Alt+key	Send Escape followed by the key, for readline shortcuts.
Backspace	Sends DEL (0x7f), which Linux programs expect. This is why backspace works in `vim`, `htop` and package managers.

Start a command directly with `Terminal -e <command>`. After the framework libraries have been rebuilt, open a *new* Terminal: an existing one keeps the old libraries until it is restarted.

8.3 File Manager

File Manager is both the file browser and the Desktop. A window has a toolbar with back, forward and up, a location bar, view controls, a directory tree and the file view. Double-click a file to open it with the handler registered in its application file; right-click for the context menu (open, open with, cut, copy, paste, rename, delete, properties, create shortcut, and for images **Set as Wallpaper**).

Desktop mode (`FileManager -d`) is the same application with a borderless window the size of the screen, showing your desktop directory as icons. In desktop mode the background context menu offers **New Directory**, **New File**, **Paste**, **Open in Terminal** and **Display Settings**.

8.4 Text Editor and the font question

Text Editor is a capable programmer's editor with syntax highlighting, line numbers, find and replace, and a configurable font. Its zoom shortcuts (**Ctrl+=** and **Ctrl+-**) change the font size only when the selected font is scalable. SDE's default is the bitmap Katica family, exactly as on SerenityOS, so out of the box the zoom does nothing.

To get scalable text, open **Display Settings ▸ Fonts** and choose a scalable family, for example **SerenitySans** or a Liberation font. The session font broker applies it to new and existing applications. This is also the first thing to try if text looks too small on a high-resolution display.

> **Bitmap fonts by default**
> SerenityOS's interface is designed around the Katica bitmap fonts. SDE keeps that default for fidelity; using a scalable font is an opt-in change.

8.5 Browser and other Internet applications

The Browser is the SerenityOS browser (the LibWeb engine). It loads HTTPS pages using the host's certificate store. Help renders the installed manual pages; Mail, Maps and Weather are the upstream applications. Assistant is a search-and-launch palette, also reachable with **Super+Space**.

Two limits are worth knowing. Certificate handling skips ECDSA root certificates, so a few sites may fail where they work in a mainstream browser. There is no WebDriver, so browser automation tools that expect it will not attach. For sites that need a mainstream engine, run Firefox or Chromium as a Linux application; they appear as normal SDE windows (Chapter 9).

8.6 System Monitor

System Monitor has four pages: **Processes**, **Performance**, **Storage** and **Network**. In SDE it reads Linux rather than SerenityOS: `/proc` for processes and CPU, `/proc/mounts` and `statvfs` for storage, and `getifaddrs` and `/proc/net` for network. The resource applets and the Taskbar open it on the relevant page.

The **Unveiled paths** tab has been removed: Linux has no equivalent of SerenityOS's unveil mechanism, and pledge/unveil are no-ops in SDE.

8.7 Developer tools

Hack Studio, Profiler, Debugger, SQL Studio and GML Playground work against Linux. The Debugger drives ptrace; the Profiler samples with `perf_event_open`, so profiling another user's processes needs `CAP_PERFMON`, and kernel samples are excluded. Crash Reporter reads Linux ELF core dumps, but it does not start automatically when an application crashes: run `CrashReporter <core>` by hand.

8.8 Games, demos and the Magnifier

The games are the upstream SerenityOS games and need no special configuration. The demos are graphics showcases; several of them are useful for checking that the compositor is rendering correctly. The **Magnifier** is an ordinary application in the Utilities menu: it shows a magnified live view of the screen under the pointer and is part of the accessibility story in Chapter 12.

8.9 Opening files and “Open with”

Opening a file uses the application handlers declared in the SerenityOS `.af` files, served by LaunchServer. The **Open with** dialog lists the handlers that declare the file's type plus Text Editor. There is not yet a full application chooser listing every installed program, and native Linux `.desktop` files are not merged into the system menu. To open a file in a specific Linux application, start that application from the Terminal or the Run dialog.

9 Settings and appearance

SDE is configured through the SerenityOS settings applications. They write plain text files under `~/.config/sde`, and the compositor reloads them live: most changes need no restart.

9.1 How settings work

Every settings application writes to a file in the SDE state directory. The compositor and the session watch those files with inotify, coalesce a burst of edits into one reload, and apply the result. An invalid value is rejected: the previous value stays in force and the reason is written to the session log. This means you can edit a file by hand while the session is running and see the effect immediately, with the same validation as the GUI.

`WindowServer.ini`	Compositor: workspaces, tiling, effects, output scale and renderer, XWayland, Taskbar grouping, accessibility, keybindings, placement, theme, fonts.
`input.ini`	Mouse and keyboard: acceleration, scrolling, tap-to-click, layouts, repeat rate.
`cursor.ini`	Cursor theme and the cursor highlight.
`screenlayout.ini`	Monitor arrangement, mode, position and scale.
`session.conf`	Session policy: idle, lock, restarts, watchdog, safe mode, autostart, restore.
`global-shortcuts.ini`	Shortcuts registered by applications through the GlobalShortcuts portal.

The effective value of a key is your file first, then the SerenityOS defaults installed with the framework. A missing key means “use the built-in default”. Chapter 15 lists every key and its default.

9.2 Display Settings

Display Settings is the main appearance and display program. It has six tabs:

Tab	What it controls
Background	The desktop colour and the wallpaper image, including how it is fitted (centre, tile, stretch, fill).
Themes	The window theme, the colour scheme and the cursor theme. Changing any of them applies to the whole desktop and to every open application, live.
Fonts	The default, fixed-width and window-title font families and sizes. Choosing a scalable font makes text zoom work in the editor.
Monitor	Resolution, position, refresh rate and scale (1×, 1.25×, 1.5×, 2×). Shows the real output and drives the compositor live.
Workspaces	The number of rows and columns in the workspace grid.
Effects	Window animations, the size/position overlay, and other compositor effects.

Multiple monitors appear as separate outputs. Drag them to arrange them, set a mode and scale for each, and pick the main screen. Hot-plugging an output applies the saved layout for that device; windows on an unplugged output move to a remaining one. **Monitor** is also where you recover from an unreadable mode: if the screen goes blank, switch to a virtual terminal and correct `screenlayout.ini`.

> **Scaling text without changing the output scale**
> Fractional output scaling scales everything, including window frames. If only text is too small, change the fonts in the Fonts tab instead: it affects the applications without changing the size of the chrome.

9.3 Mouse Settings

Mouse Settings writes `input.ini` and `cursor.ini`. Its tabs cover cursor speed and acceleration profile, scroll step and natural scrolling, double-click speed, button configuration (left-handed and swapped buttons), tap-to-click, disable-while-typing, the cursor theme and size, and the cursor highlight.

The cursor highlight is the coloured circle that **Super+H** toggles. Its colour, opacity and size are set on the Highlight tab. If you have never configured one, SDE uses SerenityOS's default: red, radius 25, 43% opacity.

9.4 Keyboard Settings

Keyboard Settings configures the layouts, variant, model and options that the compositor passes to xkb, plus the key repeat delay and rate. Changes are written to `input.ini` and applied live.

With more than one layout configured, **Alt+Shift** cycles through them. The Keymap applet shows the active layout but does not change it in this release. Compose and dead keys are not implemented yet, and there are no per-window or per-device layouts.

9.5 Themes and the Theme Editor

SDE installs the full SerenityOS theme set: Default, Dark, Light, Contrast, the Redmond family, Nord, Gruvbox Dark and a dozen more. The system menu's **Themes** submenu and **Display Settings ▸ Themes** both change the theme immediately.

A theme is a small `.ini` file describing the colours and some metrics. **Theme Editor** (Development ▸ Theme Editor) edits one, previews it live against the real widgets and saves it. Save a theme into the installed themes directory to make it appear in the theme menus. The choice is saved in `WindowServer.ini` under `[Theme]`.

A theme change repaints the background colour but leaves an image wallpaper in place, as on SerenityOS. High contrast is also an accessibility switch (§12.2).

9.6 The other settings applications

Application	What it configures
Terminal Settings	Font, bell, cursor shape and blinking, scrollback, background opacity, exit behaviour.
Network Settings	Lists the host's network interfaces and their addresses (read from Linux).
Calendar Settings	First day of the week, weekend days and the default view.
Clock Settings	12/24-hour time and the time zone.
Browser Settings	Home page, search engine, content filters, autoplay allow-list.
Mail Settings	Mail accounts and identity.
Maps Settings	Map provider and cache.
Games Settings	Game-specific preferences.
Certificate Settings	View the certificates the TLS stack trusts.
Users Settings	SerenityOS-style user accounts; on Linux this is informational, and the real accounts are Linux's.

The **Settings** item in the system menu opens the Settings launcher, which groups these programs by category.

9.7 Wallpaper

There are three ways to set the wallpaper: **Display Settings ▸ Background**, right-clicking an image in File Manager and choosing **Set as Wallpaper**, or setting `SDE_WALLPAPER` before the first start of a session. The environment variable only seeds a fresh state directory; after that, the settings file is authoritative. A bare name is looked up in the wallpaper directory; an absolute path is used as-is. `SDE_WALLPAPER_MODE` selects `center`, `tile`, `stretch` or `fill`.

The compositor paints the wallpaper before any client window, across the whole output layout; with no image, the theme's `DesktopBackground` colour is used.

9.8 Settings from the command line

In a source checkout, `sde-config` edits the same files with the compositor's parser:

```
sde-config path                          # the user configuration file
sde-config list                          # every effective value
sde-config set Output DefaultScale 1.5
sde-config validate                      # non-zero exit on an invalid value
```

The packaged install does not install `sde-config`. There, edit the file and use `/usr/lib/sde/bin/WindowServer --check-config` to validate it; Chapter 15 lists every key.

10 Linux applications, files and the clipboard

SDE is a normal Wayland desktop. Native Wayland applications run directly, X11 applications run through XWayland, and both appear as first-class windows with SerenityOS frames, Taskbar buttons and workspace membership.

10.1 Native Wayland applications

Any xdg-shell application should work: GTK 3/4, Qt 5/6, Electron, Firefox, Chromium, and terminal emulators such as foot or Alacritty. They get a SerenityOS frame unless they ask to draw their own title bar. The compositor supports the protocols these applications expect, including fractional scaling, pointer constraints for games, pointer and touch gestures, tablet and tablet-pad input, idle inhibit, keyboard-shortcut inhibit for virtual machines, presentation timing, tearing control for fullscreen video, and the system bell protocol.

Applications started from an SDE Terminal inherit `WAYLAND_DISPLAY` and `DISPLAY`, so they open on the SDE session. If you start one from outside the session, point it at the session socket explicitly; otherwise it opens on your host compositor.

10.2 X11 applications

XWayland starts with the session by default. X11 applications integrate more deeply than a bare XWayland bridge usually manages: they get SerenityOS frames, title-bar dragging and resizing, the three title-bar buttons, **Super**+drag, Taskbar buttons, Alt+Tab with thumbnails, workspace membership, window icons, the clipboard and primary-selection bridges, transient dialogs, and the usual window-state hints (always on top, skip taskbar, urgent, modal).

To turn X11 off entirely, set `[Compositor] XWayland=false` or `SDE_XWAYLAND=0`. With it off there is no X server and programs that need one will not start.

Works	Does not work yet
Frames, move/resize, maximize/minimize/close	X11 keyboard grabs (global hotkeys in X11 applications)
Taskbar, Alt+Tab, workspaces, icons	Drag and drop between X11 and Wayland applications
Clipboard and primary selection	Focus-stealing prevention
Always-on-top and skip-taskbar hints	Right-click window menu and double-click maximize on X11 frames
Transient dialogs and modal hints	Tiling shortcuts on X11 windows (use Super+drag)

> **Two X clients starting together**
> An upstream Xwayland issue can leave one of two X11 clients started at the same moment never mapped. If an X11 application does not appear, start it again; it will normally map the second time.

10.3 Starting Linux applications

There are three practical routes:

- **From the Terminal.** Type the program name. This is the most reliable way and inherits the session environment.
- **From Run… or Assistant** (**Super+Space**). Type the program name or an absolute path.

- **At login.** Add a `.desktop` file to `~/.config/autostart`. Hidden entries and `X-GNOME-Autostart-enabled=false` are honoured, and field codes such as `%U` are stripped.

Native `.desktop` entries are not merged into the SerenityOS system menu, and the menu is built from SerenityOS `.af` files. To pin a Linux application to the quick-launch area, use the Taskbar's own quick-launch management, which starts the executable directly.

10.4 Clipboard, primary selection and drag and drop

There is one clipboard. Copying in a GTK application and pasting into Text Editor works, and the clipboard history applet records text copies from either side. The X11 PRIMARY selection is bridged in both directions, so middle-click paste behaves as X11 users expect. Drag and drop works between SerenityOS applications and native Wayland applications in both directions; X11 drag and drop is not bridged yet.

10.5 Portals: file choosers, screenshots, printing and more

SDE provides an `xdg-desktop-portal` backend, `xdg-desktop-portal-serenity`. It implements Settings, FileChooser, Print, Screenshot, ScreenCast, RemoteDesktop, InputCapture, GlobalShortcuts, DynamicLauncher, AppChooser, Notification, Camera, USB, Access, Inhibit, Wallpaper, Background, Account, Location, Lockdown, Email, ProxyResolver, Realtime, PowerProfileMonitor, GameMode and MemoryMonitor. GTK and Qt applications therefore get SerenityOS file pickers and print dialogs instead of their own.

The backend is started by the session, but the SerenityOS dialogs are selected by environment variables. To see the SDE dialogs rather than the backend's fallback behaviour, export the ones you want and enable the portal routing:

```
# ~/.config/xdg-desktop-portal/portals.conf
[preferred]
org.freedesktop.portal.FileChooser=serenity
org.freedesktop.portal.Screenshot=serenity
org.freedesktop.portal.ScreenCast=serenity
org.freedesktop.portal.Print=serenity

# in the session environment
export SDE_PORTAL_FILE_CHOOSER=/usr/lib/sde/bin/FileChooserDialog
export SDE_PORTAL_SCREENSHOT_OVERLAY=/usr/lib/sde/bin/ScreenshotOverlay
export SDE_PORTAL_CHOOSER=/usr/lib/sde/bin/ScreenCastChooser
export SDE_PORTAL_PRINT_DIALOG=/usr/lib/sde/bin/PrintDialog
export SDE_PORTAL_ACCESS_DIALOG=/usr/lib/sde/bin/AccessDialog
export SDE_PORTAL_APP_CHOOSER=/usr/lib/sde/bin/AppChooserDialog
export SDE_PORTAL_COLOR_PICKER=/usr/lib/sde/bin/ColorPicker
```

Also install the portal definition and service files from `contrib/xdg-desktop-portal/` into `~/.local/share/xdg-desktop-portal/portals/` and `~/.local/share/dbus-1/services/`, and the `portals.conf` into `~/.config/xdg-desktop-portal/`. The packaged install does not do this for you yet.

> **Fallbacks without the dialog variables**
> If a dialog variable is unset, the backend does not silently do nothing: a FileChooser request fails, a ScreenCast request shares the entire screen *without asking*, an Access request is denied, USB is denied and a Print request is cancelled. Set the variables before relying on portal applications.

The Settings interface is the one most users notice: it reports the SDE theme's light/dark preference and accent colour to GTK, Qt and Flatpak applications, so they follow the SDE theme. It is also exposed over D-Bus as `org.freedesktop.portal.Settings`.

10.6 Notifications from Linux applications

The SerenityOS notification service is not a freedesktop notification daemon. Linux applications that call `org.freedesktop.Notifications` need a daemon of their own; install one such as **mako** or **dunst** and start it with the session. It will appear as an ordinary Wayland client. The portal's Notification interface can also forward to `SDE_PORTAL_NOTIFIER` or to `notify-send`.

10.7 Audio and video

Audio output goes through PipeWire. SDE's audio service serves the SerenityOS audio and mixer protocols, which is what the Audio applet, Piano and SoundPlayer use. Applications that use PipeWire or PulseAudio directly are independent of that mixer. There is no recording path through the SerenityOS audio service yet.

11 Screenshots, screen sharing and recording

The compositor supports both the legacy wlroots screencopy protocol and the portal-based capture interfaces. Which tool you use depends on whether the application is a Wayland client that speaks portals or a simple command-line utility.

11.1 Taking a screenshot

The simplest method is **grim**, which uses the wlroots screencopy protocol that SDE implements:

```
grim screenshot.png               # the whole output layout
grim -o DP-1 shot.png             # one output
grim -g "0,0 800x600" region.png  # a region
grim - | wl-copy                  # straight to the clipboard
```

On a multi-monitor session, run `grim -g` with a region, or use a portal-aware screenshot tool for an interactive selection. Applications that take screenshots through `org.freedesktop.portal.Screenshot` get the SDE screenshot overlay when `SDE_PORTAL_SCREENSHOT_OVERLAY` is set; otherwise they capture the screen directly.

> **The Screenshot application**
> The SerenityOS Screenshot application is built, but it expects SerenityOS's `/bin/shot` helper, which SDE does not provide yet. Use grim or a portal-based tool for now.

Developers can also use the session's own capture tool, `build/tools/sde-screenshot out.ppm`. It writes the raw output of the compositor and is what the automated screenshot tests use.

11.2 Sharing and recording the screen

Screen sharing and recording go through `org.freedesktop.portal.ScreenCast`, which delivers frames over PipeWire, with optional zero-copy DMA-BUF. Applications such as OBS, browsers using `getDisplayMedia`, and video-conferencing clients can capture a monitor or a window. The source selection is remembered per application.

The compositor can also provide a remote-desktop session through the RemoteDesktop portal, and input capture for applications such as remote support tools through InputCapture. Input capture is privileged and is off unless `SDE_INPUT_CAPTURE=1` is set for the compositor.

> **The screen-share chooser is opt-in**
> Without `SDE_PORTAL_CHOOSER`, a ScreenCast request is granted for the entire screen without a confirmation dialog. If you want the SerenityOS “Screen Share” chooser, point that variable at `ScreenCastChooser`.

11.3 Picking a colour

The portal's **PickColor** method is implemented. With `SDE_PORTAL_COLOR_PICKER` set, the screen freezes and a click picks the pixel colour under the pointer; without it, the backend picks a colour from a captured frame. The SerenityOS Color Picker application is also built and can be used standalone.

12 Accessibility and input methods

SDE implements the accessibility features that fit its architecture: sticky keys and high contrast in the compositor, a magnifier application, a cursor highlight, and an AT-SPI object tree that exposes SerenityOS widgets to screen readers. Text input methods work through the standard Wayland protocols.

12.1 Sticky keys

Sticky keys let you type a modifier combination one key at a time. Tap a modifier on its own and it latches: the next key press is delivered with that modifier and then clears it. Tap two modifiers to latch both. Tap the same modifier twice to cancel it.

The feature is off by default and currently has no GUI switch. Enable it in `WindowServer.ini`:

```
[Accessibility]
StickyKeys=true
```

The change is applied live.

12.2 High contrast

High contrast forces the **Contrast** palette on the compositor's window chrome and, through the theme broker, on every application. While it is on, a later theme change cannot turn it off. Enable it in the same section:

```
[Accessibility]
HighContrast=true
```

You can also simply choose the Contrast theme in **Display Settings ▸ Themes**; the accessibility switch is stronger because it pins the palette regardless of what an application requests.

12.3 Cursor highlight

Super+H toggles a coloured circle around the pointer, which makes the cursor easy to find on a busy screen or in a recording. Its colour, opacity and size are configured on the Highlight tab of Mouse Settings. The default is SerenityOS's red circle.

12.4 Magnifier

The Magnifier is in the Utilities menu. It opens a window with a magnified live view of the screen under the pointer, and it follows the pointer as you move. It is a normal application, so it can be moved to a second monitor, kept above other windows with **Always on Top**, and resized.

12.5 Screen readers and AT-SPI

SDE publishes its widget tree over AT-SPI, the Linux accessibility bus. SerenityOS widgets expose their role, name, position and state; actions such as click and toggle; readable and editable text; and focus, state and text-change events. A screen reader can therefore inspect and drive SDE applications the same way it drives GTK applications.

The session starts the AT-SPI bus helper when `at-spi2-core` is installed; set `SDE_ACCESSIBILITY=0` to opt out. Orca has not been tested with SDE, so screen-reader behaviour beyond the object model should be treated as unverified. The tool used to develop and test the tree is **accerciser**.

12.6 Keyboard layouts

Layouts are configured in Keyboard Settings or in `input.ini`, and **Alt+Shift** cycles them. The Keymap applet reports the active layout but does not switch it in this release. The layout shown by the applet comes from the session's `XKB_DEFAULT_LAYOUT`.

Compose sequences and dead keys are not implemented. For languages that need them, use an input method (below), or choose a layout that produces the characters directly.

12.7 Input methods

SDE supports `wp_text_input_v3` and `zwp_input_method_v2`, which is what modern Wayland input methods use. Start an input method inside the session:

- **fcitx5** with its Wayland frontend, for Chinese, Japanese, Korean and other IMEs.
- **maliit-keyboard** or **maliit-server**, for an on-screen keyboard.
- **squeekboard**, for phones and tablets.

Candidate windows appear under the cursor as compositor-drawn popups, and text commits work in GTK 4, Qt 6 and SerenityOS applications including Terminal. ibus-daemon does not work: it uses its own D-Bus protocol rather than text-input-v3. There is one input method per seat, and the extra text-input-v2 features are not provided.

12.8 Touch, tablets and gestures

Touchscreens, pen and tablet devices, tablet pads and switch devices are supported through libinput. Pointer constraints let games capture the pointer, and pointer gestures are forwarded to applications. Touch and tablet input can be used to drive SDE applications, although SerenityOS widgets are designed for a mouse and a keyboard first.

> **No per-device settings UI yet**
> Mouse and keyboard settings apply to the seat as a whole. Per-device profiles and per-device keyboard layouts are not exposed.

13 Differences from SerenityOS

SDE aims to be indistinguishable from SerenityOS at the level of the desktop, and deliberately different underneath. This chapter is the complete list: what is the same, what SDE adds, what behaves differently, and what is not there yet.

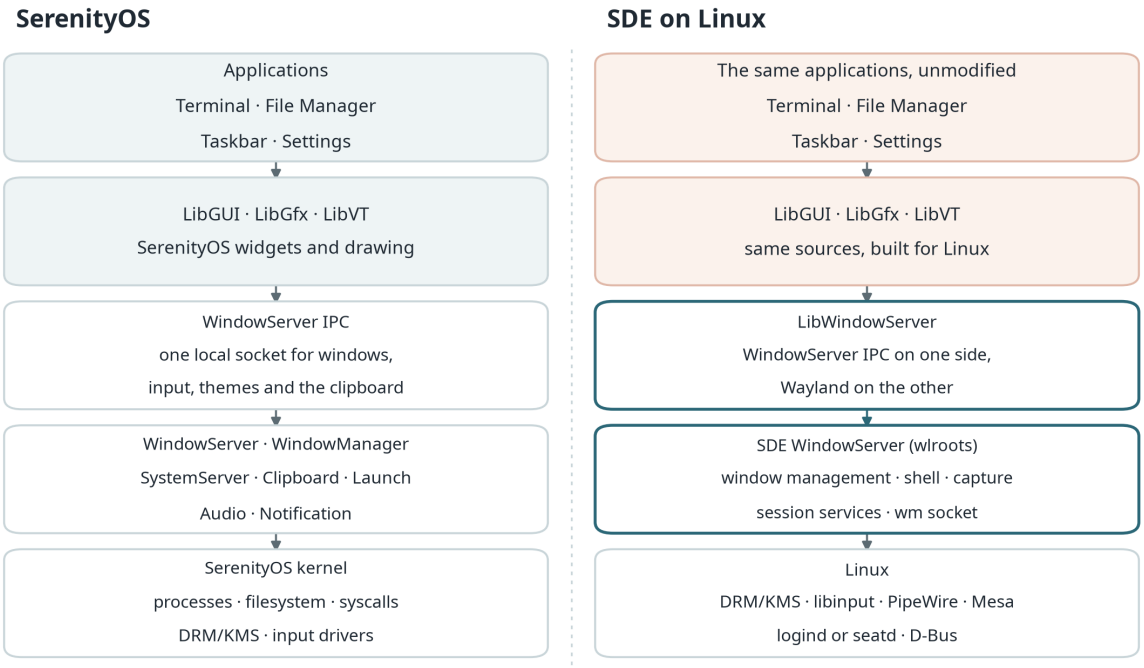
13.1 The same, on purpose

The SerenityOS applications and libraries are consumed unmodified wherever possible. That includes the Taskbar and the Desktop, the applets, the Settings applications, every application in the suite, the widget toolkit and the drawing library. A change that would need a patch is avoided if a shim at the protocol boundary can do the job instead.

In daily use this means the following are the same as SerenityOS:

- The window chrome, title-bar buttons, window menu, icon-click menu and double-click-to-maximize.
- The Taskbar's layout, system menu, quick launch, window buttons, applet area, clock and show-desktop button.
- The Alt+Tab switcher, the workspace grid and the workspace overlay.
- The application suite, its menus and dialogs, and the theme system.
- The fonts: Katica bitmap fonts by default, at the same sizes.
- The file dialogs, notification banners, clipboard history, calculator, editor, games and demos.
- The same application sources build on both systems.

13.2 Under the hood



Both columns are the same userland. Only the far side of the WindowServer protocol changes.

Figure 5. *The same userland, a different system side. In SerenityOS the WindowServer is part of the operating system. In SDE, LibWindowServer answers the application's WindowServer protocol in-process and speaks Wayland to the SDE compositor, which uses Linux for display, input, audio and the session.*

The important consequence for users is that there is no SystemServer. The session script starts the compositor, the shell, a small set of supervised services (LaunchServer, ImageDecoder, the portal backend) and then your applications. Services that applications expect are either served by a session daemon or by an in-process fallback inside the application. The “prefer the session service, else the in-process fallback” rule is what lets SDE run the same applications with or without the full set of daemons.

13.3 What SDE adds

Addition	What it gives you
Linux and Wayland	Runs on ordinary PC hardware with the Linux driver stack, from a display manager or a TTY.
Native Wayland applications	GTK, Qt, Electron, browsers, terminals and games run beside SerenityOS applications.
XWayland	X11 applications run with SerenityOS frames, Taskbar buttons, workspaces and clipboard integration.
Wi-Fi and battery applets	NetworkManager and `/sys/class/power_supply` front ends, modelled on KDE Plasma.
Portals	A complete xdg-desktop-portal backend: file chooser, print, screenshot, screen cast, remote desktop, input capture, global shortcuts and more.
Session management	Autostart, session restore, a crash watchdog, safe mode, idle blanking and locking, suspend hooks.
Accessibility	Sticky keys, high contrast, cursor highlight, the Magnifier and an AT-SPI widget tree for screen readers.
Input methods	text-input-v3 and input-method-v2, so fcitx5 and on-screen keyboards work.
Display features	Multiple outputs, hotplug, fractional scaling and per-output modes and positions.
Configuration	Validated, live-reloading text configuration and a command-line helper.
Linux audio	The SerenityOS audio protocols served by a PipeWire-backed audio service.
Multi-seat	Session and socket names are namespaced per seat.

13.4 What behaves differently

These are the differences you are most likely to notice. Some are deliberate consequences of running on Linux; some are still being brought up to parity.

Area	SerenityOS	SDE
Kernel and devices	SerenityOS kernel and drivers	Linux, DRM/KMS, libinput, Mesa, PipeWire, D-Bus
Window server	WindowServer is part of the OS	A wlroots compositor; applications reach it through LibWindowServer
System services	SystemServer owns services	No SystemServer; session daemons plus in-process fallbacks
Login and logout	LoginServer; built-in shutdown dialog actions	Your display manager; logind for power, with a fallback path
Screen locking	Built in	An external `ext-session-lock-v1` client (swaylock by default)
Fonts and zoom	Bitmap Katica; zoom is a no-op	Same default; choose a scalable font to enable text zoom
High DPI	Integer scaling	Integer and fractional scaling, with supersampled chrome at fractional scales
Window snapping	Keyboard tiling only	Keyboard tiling plus drag-to-edge snapping (an SDE addition)
Taskbar buttons	One button per window	One per window by default; optional grouping per application

Area	SerenityOS	SDE
Modal windows	Full modal blocking and “lower the blockers”	Per-application modals; cross-kind and multi-window chains are incomplete
Notifications	Server-drawn banners	Same banners, but no auto-dismiss in this release
Keyboard layouts	Switched from the Keymap applet	Switched with Alt+Shift or Keyboard Settings; the applet only reports
X11	Not applicable	Supported through XWayland, with a few gaps (§10.2)
Audio	AudioServer and AudioManager	The same protocols served over PipeWire by an SDE audio service
System Monitor	Reads SerenityOS kernel interfaces	Reads `/proc`, `statvfs` and `getifaddrs`; no Unveiled paths tab
Security model	pledge and unveil enforce sandboxing	No sandbox enforcement; pledge and unveil are no-ops
Crash reporting	Crash Reporter is part of the system	Crash Reporter reads Linux core dumps, started by hand
Home directory	`/home/anon`	Your real Linux home; Serenity configuration is seeded into `~/.config`
Resources	`/res`, `/bin`	Redirected to the installation and session directories
TLS roots	SerenityOS certificate store	The host trust store; ECDSA root certificates are skipped

13.5 Not there yet

SDE is honest about its gaps. These are known limitations in this release, not bugs in the upstream applications:

Area	Limitation
Window management	Focus-stealing prevention; maximizing across several outputs; struts beyond layer-shell; pointer tracking across several windows during one press.
Keyboard	No compose or dead keys; no per-window or per-device layouts; the WindowManagerServer keymap request is a stub; disable-while-typing and per-device profiles have no UI.
X11	No keyboard grabs, no XDND drag and drop, no right-click window menu or double-click maximize on X11 frames.
Protocols	`xdg-toplevel-drag`, `commit-timing`/`fifo` and explicit sync are not implemented.
Hardware	Multi-GPU and render-device selection, VRR, HDR, colour management, gamma and overlay planes are not implemented; the seatd and builtin seat paths are not verified on real hardware.
Applications	PixelPaint, PartitionEditor and Welcome are not built. The Screenshot app needs SerenityOS's `/bin/shot` helper.
Application launching	No full “Open with” chooser; Linux `.desktop` entries are not merged into the system menu.
Notifications	No auto-dismiss; Linux applications need a separate freedesktop notification daemon.
Portals	The SerenityOS dialogs are opt-in environment variables, and the packaged install does not install the portal definition files.
Wi-Fi	No VPN, mobile broadband or WPA-Enterprise; one icon for all security types.
Battery	No Bluetooth device batteries and no brightness control.
Recovery	Long-running applications are not supervised beyond session restore.
Packaging	The AUR packages are not published, and a clean package install and login has not yet been verified.

13.6 Where files live

SerenityOS applications refer to absolute paths such as `/res` and `/bin/Terminal`. On Linux those paths are redirected by the compatibility library, so the source code does not need to know where SDE is installed.

SerenityOS location	SDE location
`/res`	The framework's resource directory (`$SERENITY_RES_ROOT`; normally `/usr/lib/sde/share/res`), or the Serenity checkout in a development tree.
`/bin/<app>`	The session application directory (`$SDE_APP_BIN_DIR`), a symlink farm built at login.
`/etc/WindowServer.ini`	Defaults come from the framework; your values live in `~/.config/sde/WindowServer.ini`.
`/home/anon`	Your real `$HOME`.
`Base/home/anon/.config`	Seeded into `~/.config` on first start; existing files are never overwritten.
`/tmp/portal/window`	The window-manager socket (`/tmp/portal/wm`, or `/tmp/portal/wm-<seat>` on other seats).
`/tmp/session/0/portal/*`	Session service sockets for clipboard, config, launch, image, notification, audio and the rest.
Kernel interfaces	Linux equivalents: `/proc`, `/sys`, `/dev`.
`/etc/cacert.pem`	The host certificate store, through the Linux TLS integration.

13.7 What this means in daily use

- You get the SerenityOS desktop, not an imitation. The widgets, dialogs and shortcuts are the ones SerenityOS users know.
- Your Linux applications are not second-class: they get the same frames, the same Taskbar and the same workspaces.
- Tools that rely on SerenityOS-specific system behaviour (pledge/unveil, the unveiled-paths view, Crash Reporter's automatic start, the Keymap applet's layout switch) behave differently or are absent.
- The session is more Linux-like than SerenityOS-like: your display manager, logind, PipeWire, NetworkManager and portals are part of the experience.
- When in doubt about a path, a service or a shortcut, this manual's reference chapters describe SDE, not upstream SerenityOS.

14 Session, power and recovery

SDE sessions are supervised. The session script starts the compositor, restarts it if it crashes, watches its heartbeat, falls back to a minimal shell when repeated restarts fail, and records what happened in a log.

14.1 The session lifecycle

1. Your display manager (or `start-sde-session.sh`) starts the session launcher and detects the seat.
2. The launcher seeds any missing SerenityOS configuration files into `~/.config`, exports the session environment, and starts the supervised services.
3. The compositor starts with the Wayland socket, the window-manager socket and the shell startup command.
4. The shell starts the Taskbar, the Desktop and the applets. If the Taskbar fails during startup, the shell falls back to a Terminal; after startup it is restarted up to five times.
5. Once the compositor is up, the session starts the idle and sleep helpers, your XDG autostart entries and your restored applications.

A SIGTERM from the display manager is forwarded to the whole session, and the compositor and services stop cleanly. If the compositor exits with status 0, the session ends normally.

14.2 Autostart and session restore

Autostart entries are standard `.desktop` files in `$XDG_CONFIG_HOME/autostart`. Entries with `Hidden=true` or `X-GNOME-Autostart-enabled=false` are skipped, and field codes are stripped. Autostart can be disabled with `autostart=0` in `session.conf` or `SDE_SESSION_AUTOSTART=0`.

Session restore records the applications that had windows open and relaunches them after a compositor restart or at the next login. Applications that support `xdg-session-management-v1` also get their window positions back. The list is kept in `~/.config/sde/session-apps`; startup applications such as the Taskbar are skipped. Disable restore with `restore=0`.

> **A bad restored application**
> If a session loops because one restored application crashes the compositor, set `restore=0` (or start safe mode) and remove the offending entry from `session-apps`.

14.3 Locking, idle and screen blanking

Super+L locks immediately. The session runs the lock command from `lock_command`, which defaults to `swaylock`; any client implementing `ext-session-lock-v1` works. While locked, all input goes to the locker and no compositor shortcut is active.

In a native session, `sde-idle` blanks the screen after 60 seconds without input and locks after 300 seconds. Activity turns the screen back on. Both values are configurable; a nested session disables idle locking by default so it does not lock your host desktop.

```
# ~/.config/sde/session.conf
idle_seconds=300          # seconds of inactivity before locking (0 = never)
idle_dpms_seconds=60      # seconds before the screen blanks
lock_command=swaylock
```

If no lock client is installed, locking does nothing: install `swaylock` or set a command that exists. The DPMS helper can also be run by hand: `sde-dpms off` and `sde-dpms on`.

14.4 Suspend and resume

`sde-sleep.sh` watches logind's suspend signal. On suspend it locks the session (if `lock_on_suspend=1`, the default) and runs `suspend_hook`; on resume it turns the screen back on and runs `resume_hook`. Without logind or a session D-Bus it does nothing, and suspend must be triggered another way (for example `sde-power.sh suspend`).

Super+L, the idle timer and suspend all use the same locker, so the lock screen is consistent. A native SDE lock screen is planned; for now the appearance is your locker's.

14.5 Restarts, the watchdog and safe mode

Mechanism	Behaviour
Crash restart	A non-zero compositor exit is restarted after 1 second, up to 3 times. The process group is cleaned up first so no duplicate Taskbar is left behind.
Watchdog	The compositor updates a heartbeat every second. If it is older than 10 seconds, the supervisor terminates it (SIGTERM, then SIGKILL) and restarts it.
Safe mode	After the restart budget is used, or when `SDE_SAFE_MODE=1` or `safe_mode=1` is set, the session starts only the Taskbar and a Terminal: no applets, no autostart, no restore.
Give up	If safe mode itself fails three more times, the supervisor exits with an error. The log says so.
Services	LaunchServer, ImageDecoder and the portal backend are restarted up to 5 times with a backoff. On-demand services such as WebContent are not supervised.

Safe mode is the first thing to try when a configuration change or an autostart entry makes the session unusable. From a TTY, start the session with `SDE_SAFE_MODE=1 scripts/start-sde-session.sh`; from an installed session, add `safe_mode=1` to `session.conf`.

14.6 Logs and crash reports

File	Contents
`~/.local/state/sde/session.log`	The session log, when the session is started by a display manager. It records service starts, restarts, heartbeat failures, safe mode, and configuration diagnostics. It rotates at 64 MiB and keeps the previous 8 MiB as `session.log.1`.
`~/.config/sde/WindowServer-crash.log`	A backtrace written when the compositor dies from a signal (segfault, abort, bus error, illegal instruction, floating-point error). Override with `SDE_CRASH_LOG`.
`$XDG_RUNTIME_DIR/sde-heartbeat-sde`	The heartbeat file, updated every second. Useful when diagnosing a frozen compositor.
`$XDG_RUNTIME_DIR/sde-session.pid`	The supervisor's process ID; the logout fallback and the launcher use it to stop the session.

From a TTY the session log is not written; the output goes to the terminal instead. Increase detail with `[Log] Level` or `SDE_LOG_LEVEL=debug`.

15 Configuration reference

Everything SDE stores is plain text under one state directory. This chapter is the reference for those files, their keys, the environment variables that affect a session, and the command-line helpers.

15.1 The state directory

The state directory is `$SDE_STATE_DIR` if set, otherwise `$XDG_CONFIG_HOME/sde` (normally `~/.config/sde`). It contains every file in this chapter, plus the wallpaper state, the recorded session, the Xwayland display name and the crash log.

For each setting the effective value is: the environment variable if one exists, then your file, then the SerenityOS defaults installed with the framework, then the built-in default. A missing key always means "use the default". Files are watched and reloaded; invalid values are rejected and logged.

15.2 WindowServer.ini

Section	Key	Values (default)	Effect
[Workspaces]	Rows	1–16 (**2**)	Workspace grid rows.
	Columns	1–16 (**2**)	Workspace grid columns.
[Applet]	Order	comma-separated list (Serenity order + **WiFi**, **Battery**)	Applet set and order; the last entry is rightmost.
[Tiling]	EdgeThreshold	1–512 (**16**)	Distance in pixels at which drag-to-edge arms.
	DragToEdge	bool (**true**)	Enable drag-to-edge snapping.
[Focus]	FollowsMouse	bool (**false**)	Focus follows the pointer instead of clicks.
[Effects]	WindowGeometry	bool (**true**)	Show the size/position overlay while moving or resizing.
	ShowGeometry	**OnMoveAndResize**, Never, Always	When the geometry overlay appears.
	TileWindow	**ShowTileOverlay**, Never, Always	Whether the tile preview is drawn.
	AnimateWindows	bool (**true**)	Minimize, restore, launch and workspace animations.
[Output]	DefaultScale	0 = auto, or 0.5–4.0 (**0**)	Output scale, applied live; `SDE_OUTPUT_SCALE` overrides.
	Renderer	**auto**, gles2, vulkan, pixman	Compositor renderer; `SDE_RENDERER` overrides.
[Log]	Level	**error** (default), warn, info, debug	Compositor verbosity; debug is very verbose.
[Compositor]	XWayland	bool (**true**)	Run Xwayland for X11 applications.
[Taskbar]	GroupWindows	bool (**false**)	Group Taskbar buttons per application.
[Accessibility]	StickyKeys	bool (**false**)	Latch modifiers for one key press.
	HighContrast	bool (**false**)	Force the Contrast palette everywhere.
[Placement]	Windows	`title:X,Y,W,H;...`	Initial geometry rules; the first title substring match wins.
[Keybindings]	*action*	`Mod+Key[,Mod+Key]`	See §6.1 for the actions and defaults.
[Input]	AccelMin / AccelMax	**0.5** / **3.5**	Ranges shown by Mouse Settings.

Section	Key	Values (default)	Effect
	ScrollStepMin	**1**	Minimum scroll step.
	DoubleClickMin / Max	**100** / **900**	Double-click range in milliseconds.
[Fonts]	Default	**Katica 10 400 0**	Default UI font.
	FixedWidth	**Csilla 10 400 0**	Default fixed-width font.
	WindowTitle	**Katica 10 700 0**	Window title font.
[Theme]	Name / Path	**Default** / installed theme path	The active theme, written by Display Settings.

15.3 session.conf

Session policy is read from `session.conf` in the state directory (override with `SDE_SESSION_CONF`). The format is `KEY=value` with `#` comments. Environment variables take precedence over the file.

Key	Environment variable	Default
`idle_seconds`	`SDE_SESSION_IDLE_SECONDS`	300 native, 0 nested
`idle_dpms_seconds`	`SDE_SESSION_IDLE_DPMS_SECONDS`	60
`lock_command`	`SDE_LOCK_CMD`	`swaylock`
`max_restarts`	`SDE_SESSION_MAX_RESTARTS`	3
`restart_delay`	`SDE_SESSION_RESTART_DELAY`	1
`watchdog_seconds`	`SDE_SESSION_WATCHDOG_SECONDS`	10 (0 disables)
`safe_mode`	`SDE_SESSION_SAFE_MODE`	0
`autostart`	`SDE_SESSION_AUTOSTART`	1
`restore`	`SDE_SESSION_RESTORE`	1
`autostart_dir`	`SDE_SESSION_AUTOSTART_DIR`	~/.config/autostart
`lock_on_suspend`	`SDE_SUSPEND_LOCK_ON`	1
`suspend_hook`	`SDE_SUSPEND_HOOK`	(none)
`resume_hook`	`SDE_RESUME_HOOK`	(none)

15.4 input.ini, cursor.ini and screenlayout.ini

These files use SerenityOS's own formats.

File	Sections and keys
`input.ini`	**[Mouse]** Acceleration, ScrollStepSize, DoubleClickSpeed, ButtonsSwitched, NaturalScroll, AccelProfile, TapToClick, LeftHanded, DisableWhileTyping. **[Keyboard]** Layouts, Variant, Model, Options, RepeatDelay, RepeatRate.
`cursor.ini`	Cursor theme and size, and the cursor-highlight colour, opacity and size.
`screenlayout.ini`	**[Screens]** MainScreen. One **[ScreenN]** section per output with Mode, Device, Left, Top, Width, Height and ScaleFactor. The compositor writes the live layout and applies your edits.
`global-shortcuts.ini`	Shortcuts registered by applications through the GlobalShortcuts portal.

The mirror image is true as well: Display Settings, Mouse Settings and Keyboard Settings write these files through the WindowServer protocol, so a hand edit and a GUI change are the same operation.

15.5 Environment variables

Variable	Effect
SDE_STATE_DIR	Move the state directory.
SDE_WALLPAPER, SDE_WALLPAPER_MODE	Seed the wallpaper in a fresh state directory.
SDE_OUTPUT_SCALE, SDE_OUTPUT_REFRESH	Override the output scale and refresh rate.
SDE_RENDERER	Override `[Output] Renderer` (auto, gles2, vulkan, pixman).
SDE_LOG_LEVEL	Override `[Log] Level`.
SDE_XWAYLAND	Set to 0/off/false/no to disable Xwayland.
SDE_CRASH_LOG	Change the compositor crash-log path.
SDE_SESSION_LOG, SDE_SESSION_LOG_MAX_KB, SDE_SESSION_LOG_KEEP_KB	Change the session log path and rotation.
SDE_BACKENDS, SDE_ALLOW_NO_DRM	Force wlroots backends (for example headless), or allow starting without a DRM device.
SDE_SAFE_MODE	Start in safe mode.
SDE_SESSION_PORTAL	Set to 0 to skip the portal backend.
SDE_ACCESSIBILITY	Set to 0 to skip AT-SPI bus setup.
SDE_SEAT_BACKEND, LIBSEAT_BACKEND, SEATD_SOCK	Choose the seat backend.
SDE_SESSION_DISPLAY, SDE_WM_PORTAL_PATH	Change the Wayland and window-manager socket names.
SDE_SHELL_*, SDE_SERVICE_*	Tune the shell and service supervisors.
SDE_INPUT_CAPTURE	Set to 1 to allow privileged input capture (RemoteDesktop/InputCapture).
SDE_PORTAL_FILE_CHOOSER, SDE_PORTAL_CHOOSER, SDE_PORTAL_SCREENSHOT_OVERLAY, SDE_PORTAL_PRINT_DIALOG, SDE_PORTAL_ACCESS_DIALOG, SDE_PORTAL_APP_CHOOSER, SDE_PORTAL_COLOR_PICKER	Choose the SerenityOS dialog binaries used by the portal backend.
HOST_DISPLAY	The host Wayland socket for a nested session.

15.6 Command-line helpers

Command	Purpose
`sde-config path|list|get|set|validate`	Read and write the compositor configuration with validation (source checkout).
`WindowServer --print-config`	Print the effective configuration.
`WindowServer --check-config`	Validate the configuration; non-zero exit on an error.
`sde-power.sh logout|reboot|shutdown|suspend`	Session power actions, with provider detection.
`sde-capabilities [--json]`	Report the seat, logind, D-Bus, portal and power capabilities (source checkout).
`sde-dpms on|off`	Blank or unblank the outputs.
`sde-idle`	The idle daemon, normally started by the session.
`scripts/start-sde-session.sh`	Start a native session from a TTY (source checkout).
`scripts/start-sde-nested.sh`	Start a nested session inside another Wayland compositor.

15.7 A complete example

A small, self-contained `~/.config/sde/WindowServer.ini` that changes a few things and leaves the rest at their defaults:

```
[Workspaces]
Rows=2
Columns=3

[Tiling]
DragToEdge=true
EdgeThreshold=24

[Effects]
AnimateWindows=true

[Output]
DefaultScale=1.25

[Taskbar]
GroupWindows=true

[Accessibility]
StickyKeys=false
HighContrast=false

[Keybindings]
Lock=Super+L
TileLeft=Super+Left
TileRight=Super+Right

[Applet]
Order=WorkspacePicker,CPUGraph,MemoryGraph,NetworkGraph,Network,ClipboardHistory,Audio,Keymap,WiFi,Battery
```

After saving, the compositor applies the values within a moment. If a value is rejected, the session log gives the section, key and expected range.

16 Troubleshooting

Most session problems leave a clear message in one of two logs. This chapter starts there, then lists the common symptoms and their fixes, and ends with the escape hatches that get you back to a working desktop.

16.1 First steps

1. Read the session log: `~/.local/state/sde/session.log`. Look for “compositor heartbeat stale”, “restart”, “SAFE MODE”, service exits, and configuration diagnostics.
2. Read `~/.config/sde/WindowServer-crash.log` if the compositor died. A backtrace and the signal are recorded there.
3. Validate the configuration: `sde-config validate` in a checkout, or `WindowServer --check-config` in an installed tree.
4. Start in safe mode to rule out applets, autostart and restored applications: `SDE_SAFE_MODE=1` or `safe_mode=1` in `session.conf`.
5. Switch to a spare virtual terminal (**Ctrl+Alt+F2** or higher) and stop the compositor with `pkill WindowServer` if the session is unresponsive.

16.2 Common problems

Symptom	Likely cause and fix
Login returns immediately to the display manager	The compositor could not start. Check the session log for “no DRM device” or “could not create a backend”. Make sure your user can access `/dev/dri/card*` and that logind or seatd is running. As a test, set `SDE_BACKENDS=headless`.
Black screen after login, but the session is alive	A bad output mode or scale. Wait for the automatic revert; otherwise switch to a VT and correct `screenlayout.ini`, or set `[Output] DefaultScale=1`.
Screen stays black after resume	Unblank with `sde-dpms on` (installed in `/usr/lib/sde/bin`). On a packaged install the resume helper may not have been pointed at it.
Windows are slow or tearing	The GPU renderer is not in use. Check the session log for a renderer fallback; try `[Output] Renderer=gles2`, verify your Mesa driver, and reboot after a kernel or driver upgrade (an NVIDIA driver/library mismatch silently falls back to llvmpipe).
An application's window never appears	If it is an X11 application, two clients started together can hit an upstream Xwayland race: start it again. For a Wayland application, make sure `WAYLAND_DISPLAY` points at the SDE session, not at your host compositor.
No lock screen when pressing Super+L	No locker is installed, or `lock_command` is wrong. Install `swaylock` or set a command that exists.
No Wi-Fi networks	NetworkManager is not running, or the applet cannot reach the system bus. Start NetworkManager and reopen the applet. For enterprise networks, use `nmcli` or `nm-connection-editor`.
The battery applet shows a slash	The machine has no battery, or `/sys/class/power_supply` exposes none. This is normal on desktops.
The Keymap applet menu does not change the layout	Known limitation: the compositor's keymap switch is a stub. Use **Alt+Shift** or Keyboard Settings.
Text zoom does nothing in Text Editor	The default is a bitmap font. Choose a scalable font (SerenitySans or a Liberation font) in Display Settings ▸ Fonts.
A GTK application uses its own file chooser	The SDE portal dialogs are not configured. Install the portal files and set the `SDE_PORTAL_*` variables (§10.5).
File chooser requests fail in a Flatpak application	The portal backend is running but `SDE_PORTAL_FILE_CHOOSER` is unset, or the `portals.conf` is not routing FileChooser to `serenity`.
Linux applications do not send notifications	Install a freedesktop notification daemon such as mako or dunst; SDE's notification service speaks the SerenityOS protocol, not `org.freedesktop.Notifications`.
Alt+Escape ended the session	This is a fixed development shortcut and cannot be disabled. Avoid it; session restore brings your applications back.
The Taskbar disappeared	The shell restarts it up to five times; if it keeps failing, the shell falls back to a Terminal. Check the session log. In safe mode you get a Taskbar and a Terminal without applets.

16.3 Escape hatches

Situation	What to do
The desktop is unusable	Switch to a VT (**Ctrl+Alt+F2**) and `pkill WindowServer`; the session supervisor will restart it, or log in again.
A configuration change broke the session	Set `SDE_SAFE_MODE=1`, then fix or delete the offending file.
An autostart entry or a restored app is the problem	Start with `autostart=0` or `restore=0` in `session.conf`.
The GPU path is broken	Set `SDE_RENDERER=pixman` or `[Output] Renderer=pixman`.
X11 breaks the session	Set `SDE_XWAYLAND=0`.
You need detail	Set `SDE_LOG_LEVEL=debug` and reproduce; read the session log.
Remove the development session	`sudo scripts/install-session.sh --uninstall`.

16.4 Reporting a problem

A useful report includes: what you did, what you expected, what happened; the session log and the crash log (redact anything private); your distribution, GPU and driver; whether the session is native, nested or headless; and your `WindowServer.ini` and `session.conf` if they differ from the defaults. If the problem is visual, a screenshot taken with grim is worth more than a description.

> **Before you report a limitation**
> Check §13.5. Many surprising behaviours are known gaps with a documented reason, not regressions.

A Shortcut card

The bindings you are most likely to want, on one page. The configurable ones live under `[Keybindings]`.

Caution: Alt+Escape quits the compositor immediately and cannot be disabled.

Windows

Shortcut	Action
Super+← / **Super+→**	Tile left / right
Super+↑ / **Super+↓**	Maximize / minimize
Super+Alt+arrows	Maximize horizontally or vertically
Super+left-drag	Move a window
Super+right-drag	Resize a window
Double-click title bar	Maximize / restore
Right-click title bar	Window menu
Super+H	Cursor highlight

Workspaces and switching

Shortcut	Action
Ctrl+Alt+arrows	Switch workspace
Ctrl+Shift+Alt+arrows	Carry the focused window to another workspace
Alt+Tab / **Shift+Alt+Tab**	Switch windows on this workspace
Super+Tab	Switch windows across all workspaces
Super+D	Show desktop / restore

Session

Shortcut	Action
Super (tap)	System menu
Super+Space	Assistant
Super+1…9	Taskbar window button 1–9
Super+L	Lock
Alt+Shift	Next keyboard layout
Alt+Escape	Quit the session immediately (not configurable)

Window menu and Terminal

Keys	Action
Esc	Close an open window menu
↑ / ↓	Move through the window menu
Enter	Activate the highlighted item
Ctrl+Shift+C / **Ctrl+Shift+V**	Copy / paste in Terminal
Ctrl+C	Interrupt the foreground program in Terminal

B Glossary

AT-SPI	The Linux accessibility bus and object model. SDE publishes its widget tree over it so screen readers can inspect applications.
Compositor	The program that owns the display, receives input, decides where windows go and combines their contents. SDE's compositor is **WindowServer**.
DRM/KMS	The Linux kernel's display subsystem: it sets modes and gives a process permission to drive the screen (DRM master).
ext-session-lock-v1	The Wayland protocol for a lock screen that stays up even if the session crashes. SDE delegates locking to a client such as swaylock.
Font broker	The session service that tells applications which theme and fonts are active, so they change without a restart.
Layer-shell	The Wayland protocol used for panels and overlays. The Taskbar is a layer-shell client.
LibGUI / LibGfx	SerenityOS's widget toolkit and drawing library. SDE runs them unmodified on Linux.
LibWindowServer	The compatibility library that implements the SerenityOS WindowServer protocol in-process and translates it to Wayland.
libinput	The Linux library that reads input devices (keyboards, mice, touchpads, tablets, touchscreens).
PipeWire	The Linux audio and video server used for sound and for portal screen capture.
Portal	A D-Bus interface through which sandboxed or desktop-agnostic applications ask the desktop for files, screenshots, screen sharing and similar services.
Seat	A set of input and output devices belonging to one user session. logind or seatd grants access to it.
Session restore	Relaunching the applications that had windows open when the session ended or the compositor restarted.
Wayland	The display protocol SDE speaks. Applications are Wayland clients of the compositor.
wlroots	The compositor library SDE builds on. It provides the backend, renderer and protocol implementations.
XWayland	An X11 server that runs as a Wayland client, so X11 applications can appear in a Wayland session.

C Further reading and license

SDE is documented in depth in the project's `docs/` directory. The following documents are the most useful companions to this manual:

Document	Why read it
`docs/01-objectives.md`	Vision, goals, non-goals and success criteria.
`docs/04-known-gaps.md`	The running list of gaps and their current state, in more detail than Chapter 13.
`docs/14-r4.4-config.md`	The configuration schema and live-reload design.
`docs/23-wayland-protocols.md`	Every Wayland protocol the compositor advertises.
`docs/29-accessibility.md`	Accessibility design, including sticky keys and AT-SPI.
`docs/31-compositor-design.md`	The compositor's architecture, for contributors.
`docs/26-ci.md`	How SDE is tested, and what a test machine can and cannot cover.

The source repositories are:

- **SDE** — the compositor, session and services described here.
- **libgui-wayland** — LibWindowServer and the SerenityOS/Linux compatibility layer.
- **SerenityOS** — the upstream source of the shell, applications, widgets, themes and fonts.
- **wlroots** — the compositor library.

License

SDE is distributed under the BSD 2-Clause License, the same license as SerenityOS, whose unmodified sources it builds on. The full text is in the `LICENSE` file in the repository.

> **A love letter, twice**
> SerenityOS describes itself as “a love letter to '90s user interfaces”. SDE is a love letter to SerenityOS: the desktop it built, carried onto the Linux system that runs on modern hardware.